

MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 9: Parabolic Equations, Implicit Schemes, and the Grid Péclet Number

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1 Introduction

Up to this point, our numerical treatment has focused primarily on pure hyperbolic transport (linear advection $u_t + au_x = 0$). However, real fluid flows governed by the Navier-Stokes equations invariably combine convective transport with dissipative viscous friction. Viscous stresses introduce second-order spatial derivatives, giving rise to parabolic and mixed **advection-diffusion equations**.

In this lecture, we complete Module 2 by examining:

1. The severe stability bottleneck of explicit methods for diffusion ($u_t = \alpha u_{xx}$) and the motivation for **implicit time-stepping**.
2. Unconditionally stable implicit schemes: **Backward-Time Central-Space (BTCS)** and the second-order **Crank-Nicolson** method.
3. Efficient inversion of tridiagonal linear systems via the **Thomas Algorithm (TDMA)**.
4. The coupled **advection-diffusion equation** ($u_t + au_x = \alpha u_{xx}$) and the crucial role of the **Grid Péclet number** Pe_Δ .

2 The Pure Diffusion Equation

Consider the 1D heat / diffusion equation:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \quad \alpha > 0 \quad (1)$$

2.1 Explicit FTCS Scheme and the Parabolic Time-Step Barrier

Approximating the time derivative by forward Euler and the second spatial derivative by central differences:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \alpha \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \quad (2)$$

Defining the dimensionless **diffusion number** (mesh Fourier number):

$$r = \frac{\alpha \Delta t}{\Delta x^2} \quad (3)$$

The explicit update formula is:

$$u_j^{n+1} = r u_{j-1}^n + (1 - 2r) u_j^n + r u_{j+1}^n \quad (4)$$

2.1.1 Von Neumann Stability Analysis of Explicit Diffusion

Substituting the Fourier mode $u_j^n = \hat{u}^n e^{ij\xi}$ with phase angle $\xi = k\Delta x$:

$$\begin{aligned} G(r, \xi) &= 1 - 2r + r(e^{i\xi} + e^{-i\xi}) = 1 - 2r + 2r \cos \xi \\ &= 1 - 2r(1 - \cos \xi) = \boxed{1 - 4r \sin^2 \left(\frac{\xi}{2} \right)} \end{aligned} \quad (5)$$

To satisfy $|G| \leq 1$ for all $\xi \in [-\pi, \pi]$:

$$-1 \leq 1 - 4r \sin^2(\xi/2) \leq 1 \quad (6)$$

The upper bound $1 - 4r \sin^2(\xi/2) \leq 1$ holds trivially since $r > 0$. The lower bound requires:

$$1 - 4r \geq -1 \implies 4r \leq 2 \implies r = \frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2} \quad (7)$$

The Parabolic Bottleneck: $\Delta t \sim \Delta x^2$

Unlike hyperbolic advection where $\Delta t \leq \frac{\Delta x}{a}$ scales linearly with Δx , parabolic diffusion requires:

$$\Delta t \leq \frac{\Delta x^2}{2\alpha}$$

If the spatial grid is refined by a factor of 10 ($\Delta x \rightarrow \Delta x/10$), the time step must be cut by a factor of ****100****! For fine computational meshes, explicit time-stepping becomes computationally prohibitive.

3 Implicit Time-Stepping Methods

To eliminate the restrictive time-step constraint $r \leq 1/2$, we evaluate spatial derivatives at the ****new time level**** t^{n+1} .

3.1 Backward-Time Central-Space (BTCS)

Evaluating the second derivative at $n + 1$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \alpha \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2} \quad (8)$$

In terms of $r = \frac{\alpha \Delta t}{\Delta x^2}$, grouping unknown values at level $n + 1$ on the left:

$$-r u_{j-1}^{n+1} + (1 + 2r) u_j^{n+1} - r u_{j+1}^{n+1} = u_j^n \quad (9)$$

3.1.1 Stability of BTCS

Substituting $u_j^n = \hat{u}^n e^{ij\xi}$:

$$\hat{u}^{n+1} [(1 + 2r) - 2r \cos \xi] = \hat{u}^n \implies G_{\text{BTCS}}(\xi) = \frac{1}{1 + 4r \sin^2(\xi/2)} \quad (10)$$

Because $r > 0$ and $\sin^2(\xi/2) \geq 0$, the denominator is strictly ≥ 1 for all ξ . Hence:

$$|G_{\text{BTCS}}(\xi)| = \frac{1}{1 + 4r \sin^2(\xi/2)} \leq 1 \quad \forall r > 0, \quad \forall \xi \in [-\pi, \pi] \quad (11)$$

BTCS is unconditionally stable for any time step $\Delta t > 0$. However, it is only first-order accurate in time, $\mathcal{O}(\Delta t, \Delta x^2)$.

3.2 The Crank-Nicolson Scheme (Second-Order Accurate)

In 1947, John Crank and Phyllis Nicolson averaged the spatial derivative between time levels n and $n + 1$ (trapezoidal rule):

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\alpha}{2} \left[\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2} + \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \right] \quad (12)$$

Rearranging:

$$-\frac{r}{2}u_{j-1}^{n+1} + (1+r)u_j^{n+1} - \frac{r}{2}u_{j+1}^{n+1} = \frac{r}{2}u_{j-1}^n + (1-r)u_j^n + \frac{r}{2}u_{j+1}^n \quad (13)$$

- **Order of accuracy:** $\mathcal{O}(\Delta t^2, \Delta x^2)$ (second-order in both space and time).
- **Amplification factor:**

$$G_{\text{CN}}(\xi) = \frac{1 - 2r \sin^2(\xi/2)}{1 + 2r \sin^2(\xi/2)} \implies |G_{\text{CN}}(\xi)| \leq 1 \quad \forall r > 0 \quad (14)$$

Crank-Nicolson is **unconditionally stable** (A-stable).

3.3 Solving the Linear System: The Thomas Algorithm (TDMA)

Both BTCS and Crank-Nicolson lead to a coupled **tridiagonal linear system** $\mathbf{A}\mathbf{u}^{n+1} = \mathbf{d}$ at every time step:

$$a_j u_{j-1}^{n+1} + b_j u_j^{n+1} + c_j u_{j+1}^{n+1} = d_j, \quad j = 1, \dots, N-1 \quad (15)$$

Because the matrix is tridiagonal and strictly diagonally dominant ($|b_j| > |a_j| + |c_j|$), it can be solved directly in $\mathcal{O}(N)$ arithmetic operations via Gaussian elimination without pivoting, known as the **Thomas Algorithm** (Tridiagonal Matrix Algorithm, TDMA).

4 The Advection-Diffusion Equation and the Grid Péclet Number

When fluid flows with convection speed a and kinematic viscosity / diffusivity α :

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \alpha \frac{\partial^2 u}{\partial x^2} \quad (16)$$

4.1 The Grid Péclet Number Pe_Δ

The ratio of convective transport to diffusive transport over a single grid cell is given by the dimensionless **Grid Péclet Number**:

$$Pe_\Delta = \frac{a \Delta x}{\alpha} \quad (17)$$

4.2 The Danger of Central Differencing Convection ($Pe_\Delta > 2$)

Consider the steady 1D boundary-layer problem $au_x = \alpha u_{xx}$ on $[0, 1]$ with $u(0) = 0$ and $u(1) = 1$. Discretizing convection with central differences:

$$a \frac{u_{j+1} - u_{j-1}}{2\Delta x} = \alpha \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} \quad (18)$$

Rearranging in terms of $Pe_\Delta = \frac{a\Delta x}{\alpha}$:

$$-(1 + Pe_\Delta/2)u_{j-1} + 2u_j - (1 - Pe_\Delta/2)u_{j+1} = 0 \quad (19)$$

The characteristic roots of this discrete difference recurrence are:

$$\rho_1 = 1, \quad \rho_2 = \frac{1 + Pe_\Delta/2}{1 - Pe_\Delta/2} \quad (20)$$

- When $Pe_\Delta \leq 2$: $\rho_2 > 0$, the discrete solution is smooth and monotonic.
- When $Pe_\Delta > 2$: $1 - Pe_\Delta/2 < 0 \implies \rho_2 < 0$! The root becomes **negative**, which generates an unphysical alternating factor $(-\rho_2)^j = (-1)^j |\rho_2|^j$.

As shown in Figure 1 (Right), whenever the grid cannot resolve the viscous boundary layer ($Pe_\Delta > 2$), central differencing generates **catastrophic spatial oscillations**. Upwind differencing replaces $(1 - Pe_\Delta/2)$ with positive coefficients, completely curing the wiggles at the expense of introducing $\mathcal{O}(\Delta x)$ numerical diffusion.

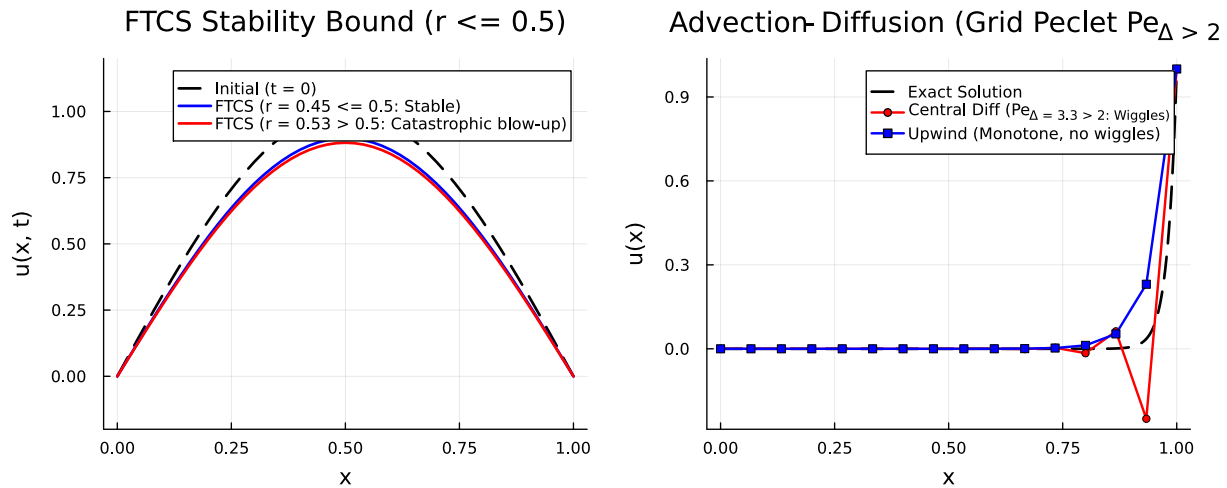


Figure 1: Left: FTCS numerical instability for heat equation when $r > 0.5$. Right: Steady advection-diffusion boundary layer at $Pe_\Delta = 3.3 > 2$: central difference produces spurious wiggles, whereas upwind differencing guarantees strict monotonicity. Generated with Julia.

5 Summary of Finite Difference Schemes for Flow Equations

Equation	Scheme	Type	Order	Stability Criterion
Diffusion ($u_t = \alpha u_{xx}$)	FTCS	Explicit	$\mathcal{O}(\Delta t, \Delta x^2)$	$r = \frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2}$
Diffusion	BTCS	Implicit	$\mathcal{O}(\Delta t, \Delta x^2)$	Unconditionally Stable
Diffusion	Crank-Nicolson	Implicit	$\mathcal{O}(\Delta t^2, \Delta x^2)$	Unconditionally Stable
Advection ($u_t + au_x = 0$)	Upwind (FOU)	Explicit	$\mathcal{O}(\Delta t, \Delta x)$	$\nu = \frac{a \Delta t}{\Delta x} \leq 1$
Advection	Lax-Wendroff	Explicit	$\mathcal{O}(\Delta t^2, \Delta x^2)$	$ \nu \leq 1$
Adv.-Diffusion	Central	Explicit	$\mathcal{O}(\Delta t, \Delta x^2)$	Monotone for $Pe_\Delta \leq 2$

6 Exercises

Exercises for Lecture 9

E1. Crank-Nicolson Derivation: Starting from equation (13), apply the discrete Fourier ansatz $u_j^n = \hat{u}^n e^{ij\xi}$ and prove in detail that:

$$G_{\text{CN}}(\xi) = \frac{1 - 2r \sin^2(\xi/2)}{1 + 2r \sin^2(\xi/2)}$$

Show that $|G_{\text{CN}}(\xi)| \leq 1$ for all $r > 0$ and all ξ . Under what condition does $G_{\text{CN}}(\xi) \rightarrow -1$ (leading to non-physical high-frequency ringing)?

E2. Thomas Algorithm Complexity: Write down pseudocode for the Thomas Algorithm to solve $\mathbf{A}\mathbf{u} = \mathbf{d}$. Count the total number of floating-point multiplications and additions, and compare its efficiency with standard $\mathcal{O}(N^3)$ Gaussian elimination for $N = 10^4$ grid points.

E3. Upwind Boundary Layer Resolution: Discretize the 1D steady advection-diffusion equation $au_x = \alpha u_{xx}$ using upwind differencing for convection: $a \frac{u_j - u_{j-1}}{\Delta x} = \alpha \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$. Show that the discrete characteristic root is $\rho = 1 + Pe_\Delta > 0$, proving that the upwind solution is monotonically increasing for all values of $Pe_\Delta > 0$.

References

- [1] John D. Anderson. *Computational Fluid Dynamics: The Basics with Applications*. McGraw-Hill, New York, NY, 1995.
- [2] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press, Cambridge, UK, 2002.